

PTVA's MULUND COLLEGE OF COMMERCE (AUTONOMOUS)			
Programme: B.Sc. Information Technology		Class: TY	Semester: V (AUTO)
Exam: ESE (ATKT)	Name of the Course: Big Data and NOSQL		
Date of Exam: 01/12/2025	Marks: 60	Time: 2 hours	Autonomous
Instructions: (1) <u>All</u> questions are <u>compulsory</u>. (2) Make <u>suitable assumptions</u> wherever necessary and <u>state the assumptions</u> made. (3) Answers to the <u>same question</u> must be <u>written together</u>. (4) Numbers to the <u>right</u> indicate <u>marks</u>. (5) Draw <u>neat labeled diagrams</u> wherever <u>necessary</u>. (6) Use of non-programmable calculator is <u>allowed</u>.			

Q 1 Attempt any TWO of the following: 12

- A. Why is it important to understand the definition of Big Data?
- B. Explain the importance of Big Data in context to its usage.
- C. Describe what the CAP theorem states?
- D. Write a short note on Non-Relational Approach.

Q 2 Attempt any TWO of the following: 12

- A. Explain the purpose of each tool available in MongoDB?
- B. List a set of guidelines for data import in shared environments.
- C. Can you explain what polymorphic schema means in database design?
- D. What factors influence the election of a primary member in a MongoDB replica set and how do they interact?

Q 3 Attempt any TWO of the following: 12

- A. What are some best practices for coding with MongoDB?
- B. How do different storage engines affect performance and data management in MongoDB?
- C. What are some limitations of sharding in MongoDB?
- D. Can you describe how MongoDB's default configurations may pose security risks?

Q 4 Attempt any TWO of the following: 12

- A. What method is used in jQuery to add a CSS class to an element?
- B. Why is it important to understand DOM manipulation when developing web applications?
- C. What are the benefits of using TimesTen over traditional database systems?
- D. Can you draw a diagram that represents the Spark architecture, including the flow of data between components?

Q 5 Attempt any TWO of the following: 12

- A. What are the advantages and disadvantages of using JSON over XML for data interchange?
- B. Why is it important to adhere to JSON grammar rules in data interchange?
- C. How would you use JSON.stringify() to prepare an object for sending to a server?
- D. JSON data can be made persistent. Justify.